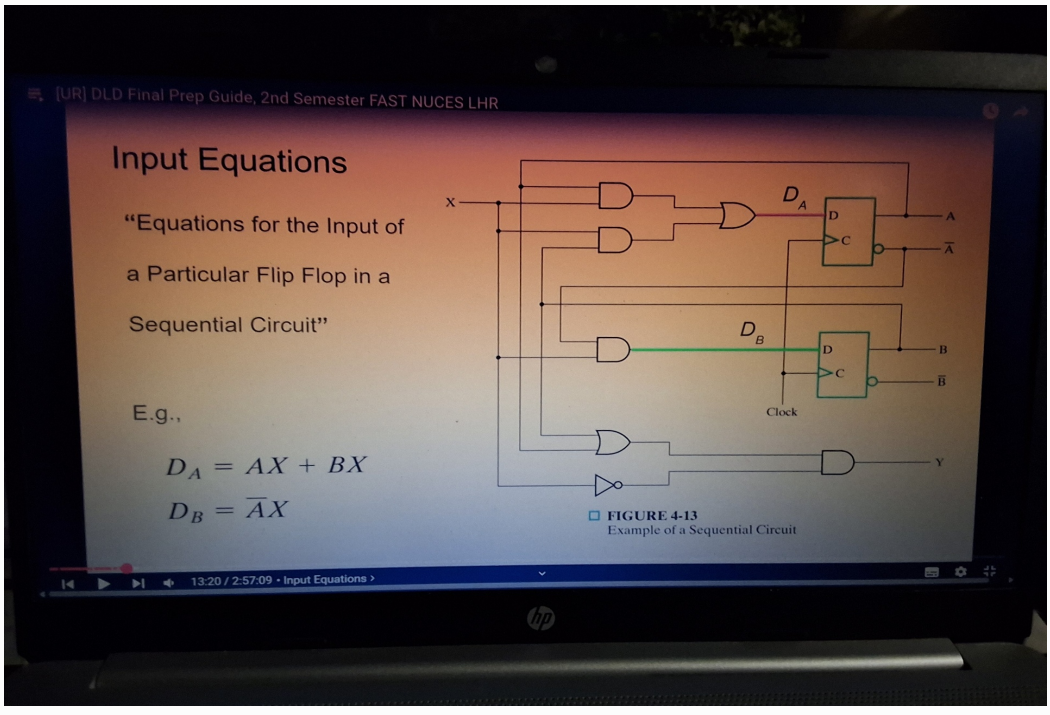
DLD

Sequential circuit Analysis



Synchronous

→ easy to design

Same clocks

fast

Asynchronous counters

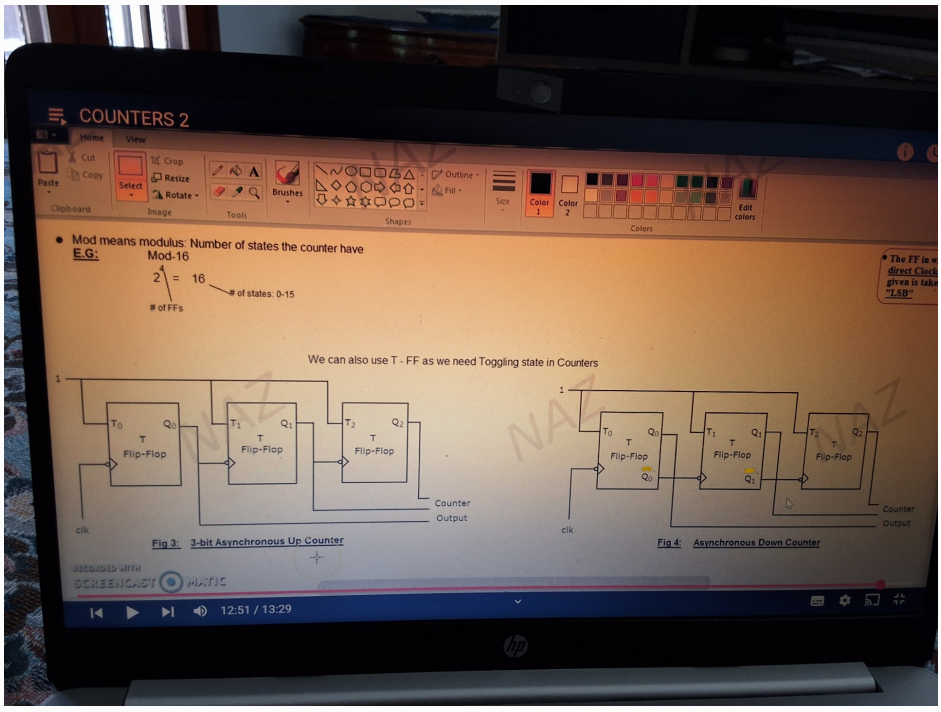
Difficult

Diff clock for all FF.

Slow.

A counter can count from 2 to b | 0 to 10

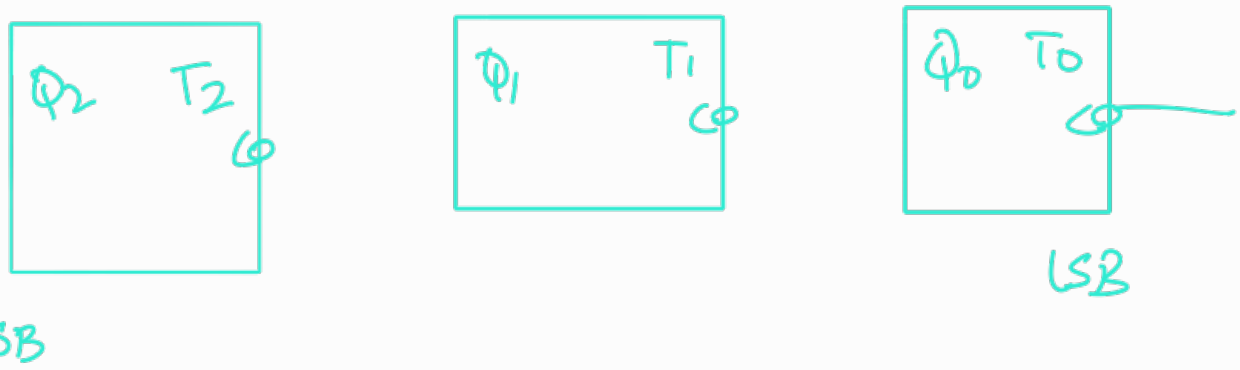
depends on design.



Make a Mod 6 Ripple up counter. (0-5)

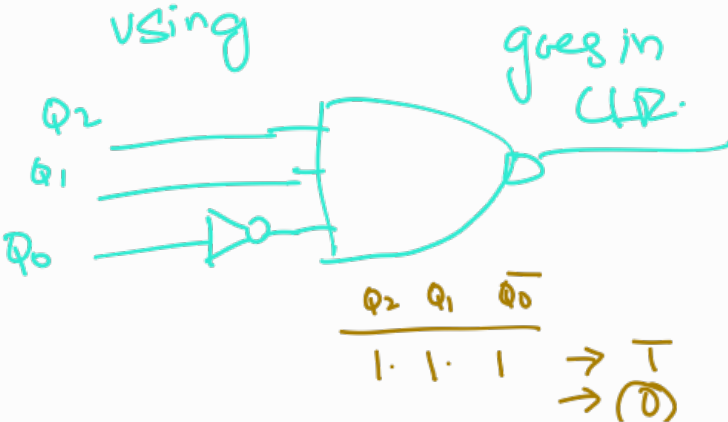
using $2^3 = 8$ mod. to count from (0-7)

(Toggle FF)

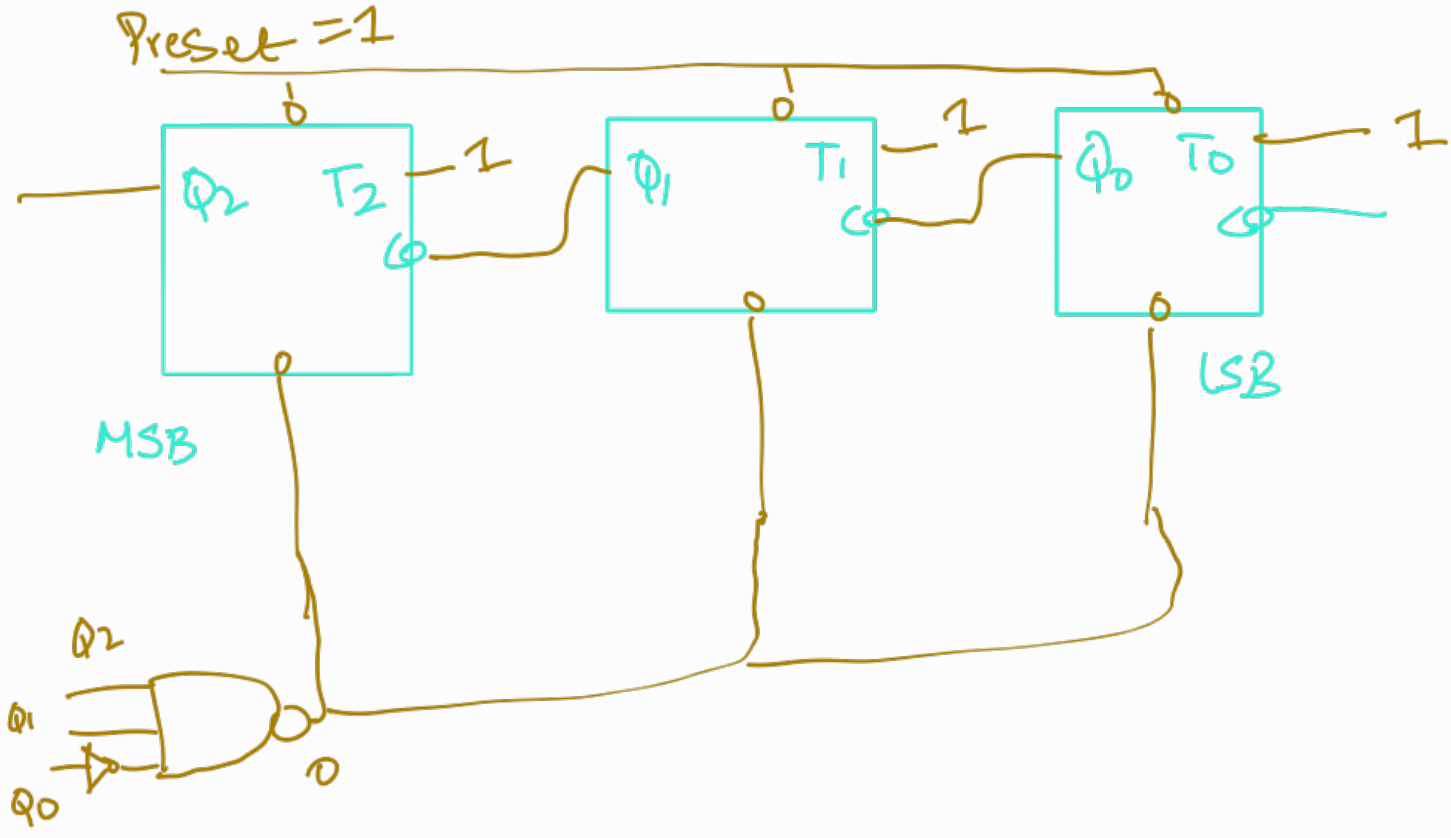


Q_2	Q_1	Q_0
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1

(Possible States)

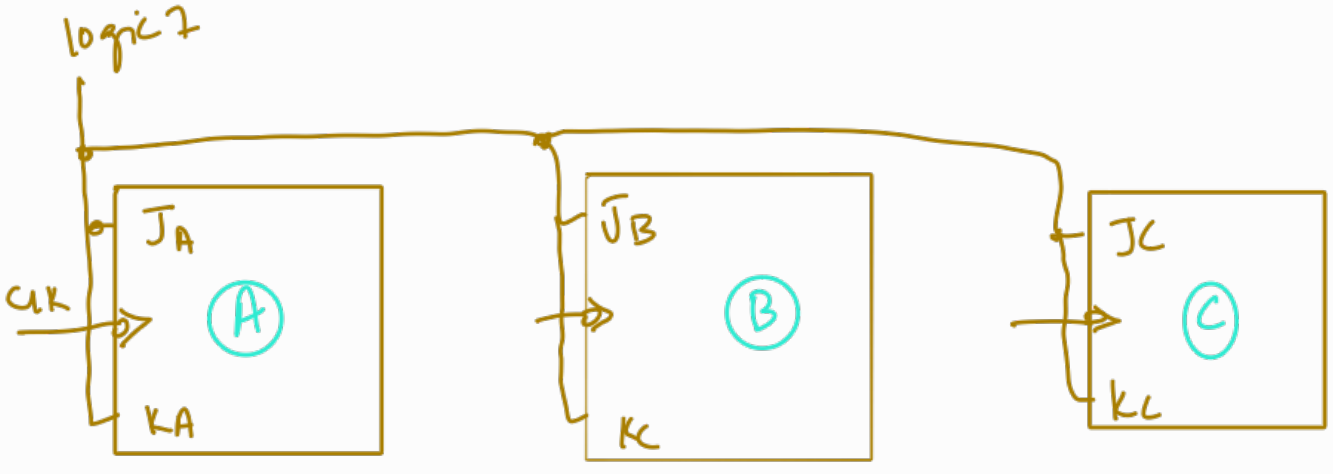


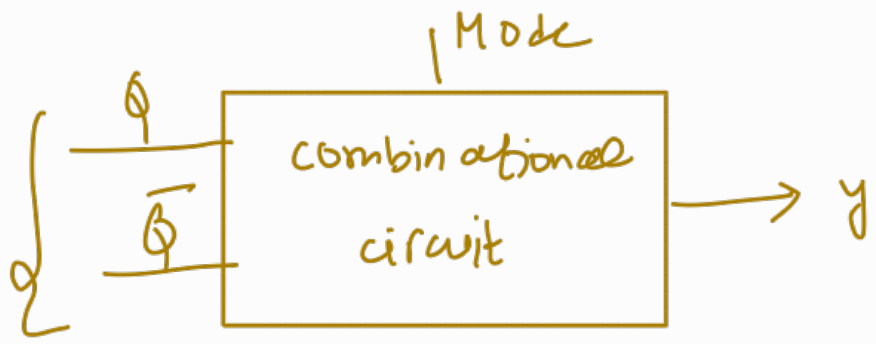
(1 1 0)
(1 1 1) } Ignore goes back to zero.



3 bit UP/ down (Ripple) counter:

A mode control input(M) is used to select either up or down mode





output of previous ff

Q connected.

for M=0 up counting.

M=1 Down counting.

$\bar{Q}$ connected to CLK

M	Q	$\bar{Q}$	y
0	0	1	0
0	0	1	0
0	1	0	1
0	1	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1

M=0 → copies Q

M=1 → copies $\bar{Q}$

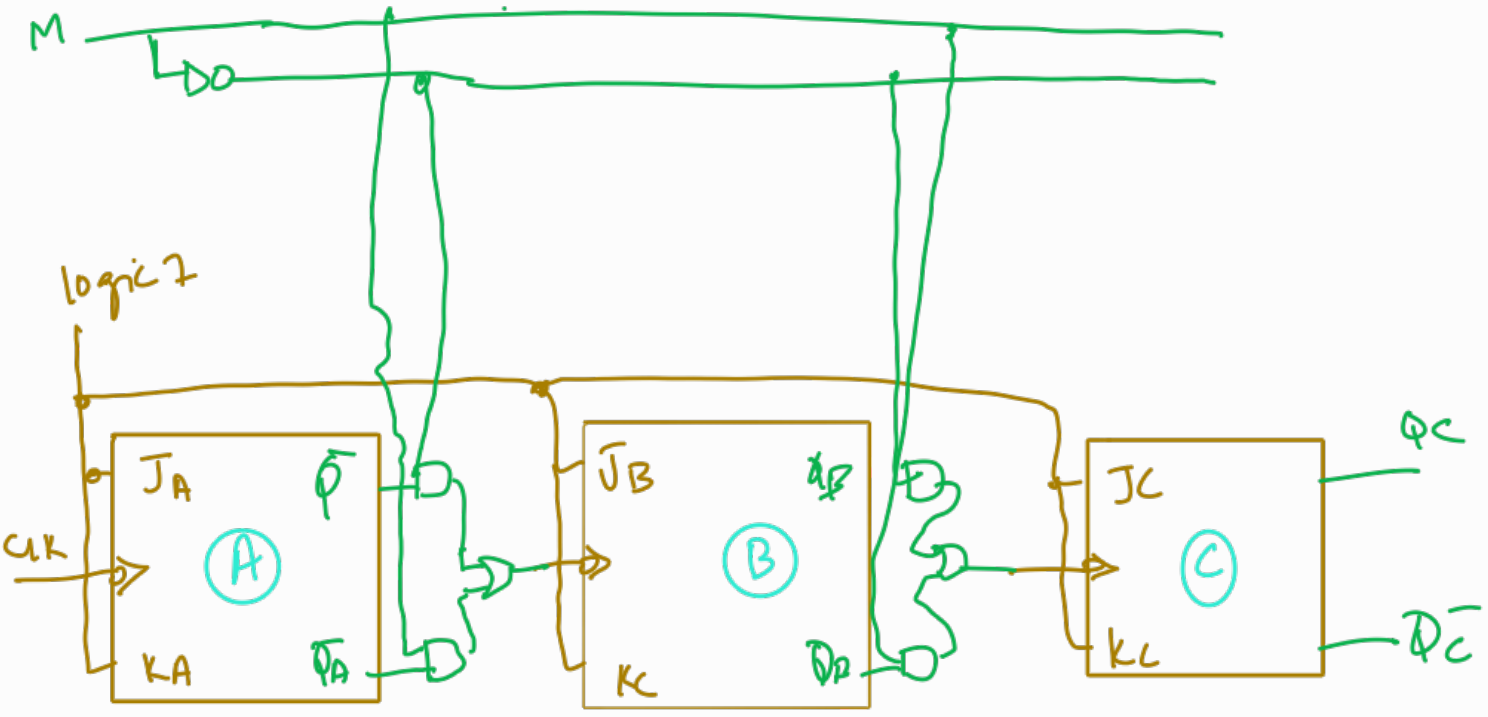
Making y kmap:

M \ Q	00	01	11	10
0	0	0	1	1
1	0	1	1	0

I, II

$y = I + II$

$y = \bar{M}Q + M\bar{Q}$



Design a 3bit Synchronous up counter using DFF.

How to design Synchronous counters.

1) Decide the no. of FF.
2) excitation table of FF
3) State diagram and circuit table
4) Simplified K-map eqn
5) Draw logic diagram.

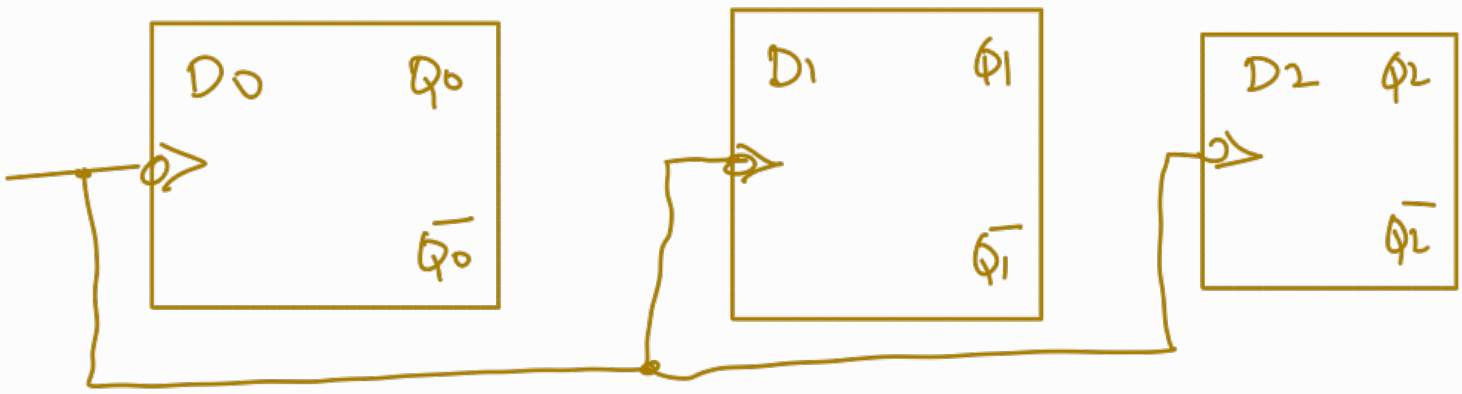
Sol:: 3 ff needed. $2^3 = (0-7)$ will count

using D ff Excitation table

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

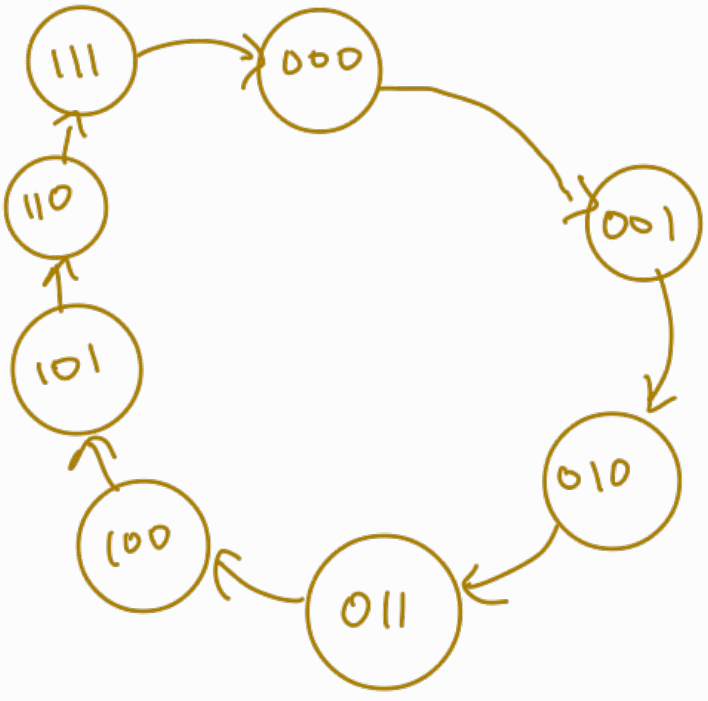
LSB

MSB



Clock

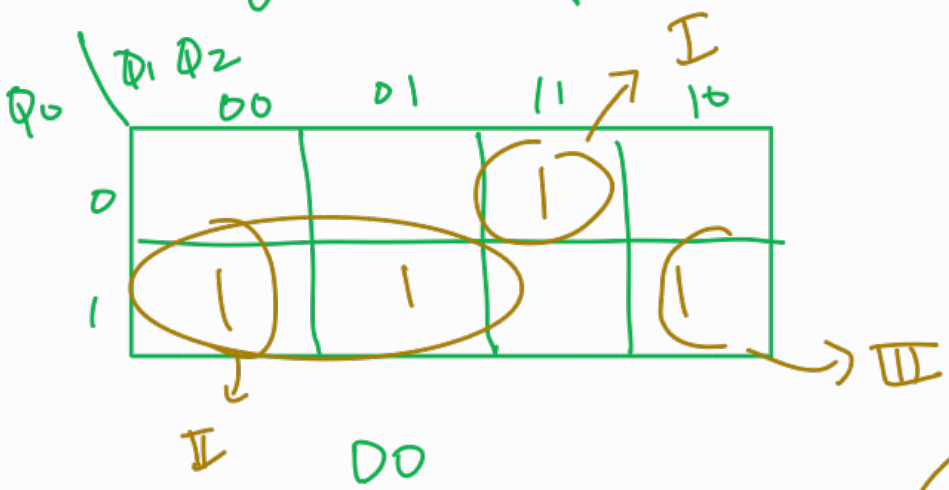
State diagram



Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Q_0	Q_1	Q_2	D_0	D_1	D_2
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0

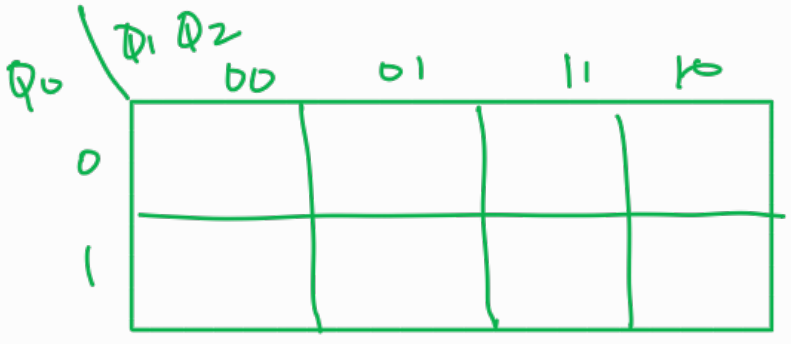
Making 3 K-maps



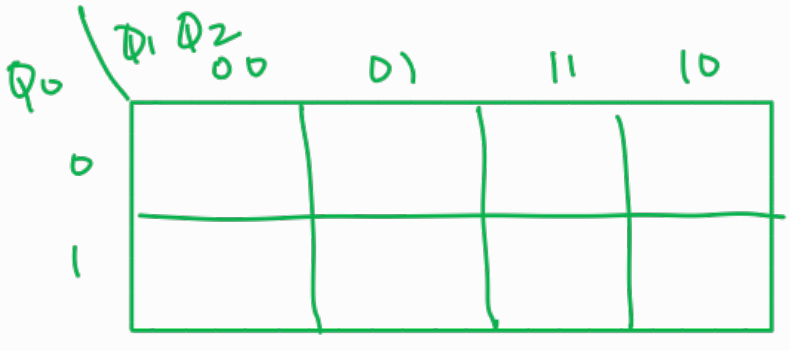
Mapping ones.

D_0 = $\overline{Q_0}\,Q_1\,Q_2 + Q_0\,\overline{Q_2} + Q_0\,\overline{Q_1}$

DO



$D_1 = Q_0 \oplus Q_1$



$D_2 = \overline{Q_0}$

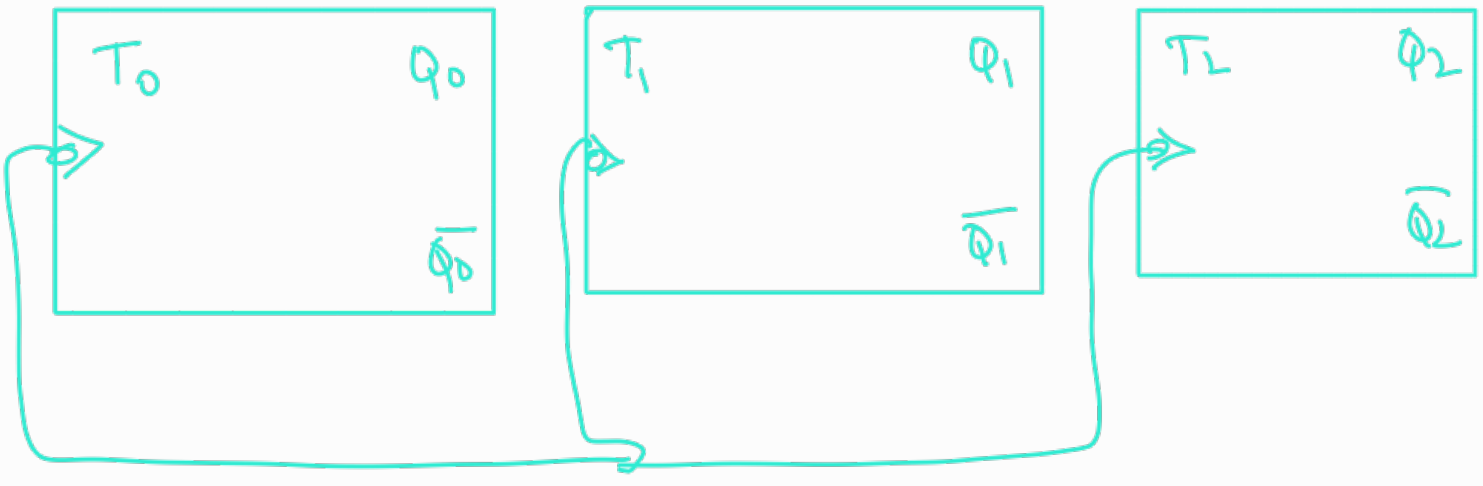
Now following this make the circuit diagram.

3 bit up/down Synchronous counter:

Important note:. in Synchronous counter you need excitation table of FF.
But in Aysenchronous counter there is no need as each ff toggles based on previous FF one

now using t flip flop.

$2^3 = (0-7)$



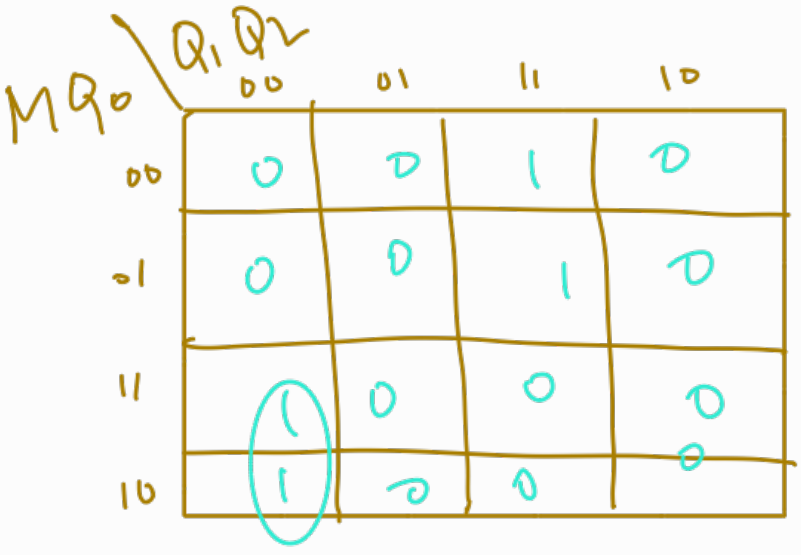
TFF excitation table:

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

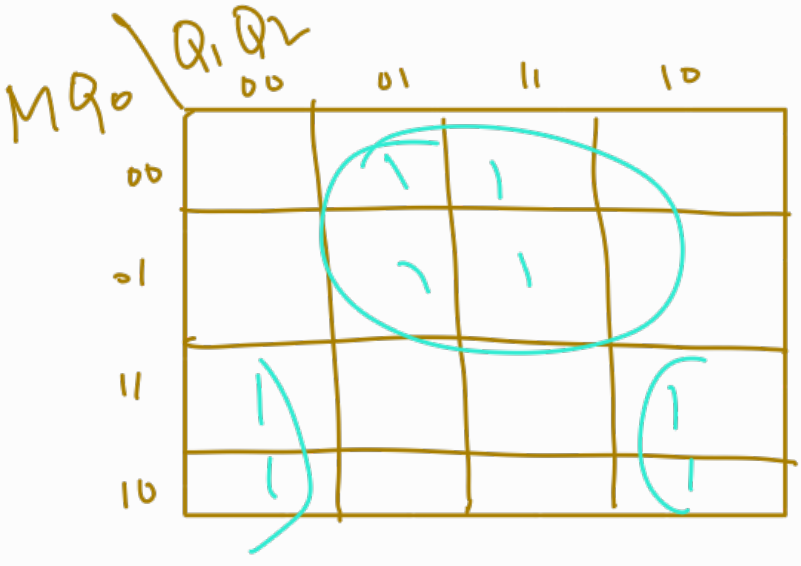
M	Q_0	Q_1	Q_2	T_0	T_1	T_2
0	0	0	0	0	0	1
0	0	0	1	0	1	1
0	0	1	0	0	0	1
0	0	1	1	1	1	1
0	1	0	0	0	0	1
0	1	0	1	0	1	1
0	1	1	0	0	0	1
0	1	1	1	1	1	1
1	0	0	0	1	1	1
1	0	0	1	0	0	1
1	0	1	0	0	1	1
1	0	1	1	0	0	1
1	1	0	0	1	1	1
1	1	0	1	0	0	1
1	1	1	0	0	1	1
1	1	1	1	0	0	1

M=0 Up M=1 Down

Finding Minimized expressions.

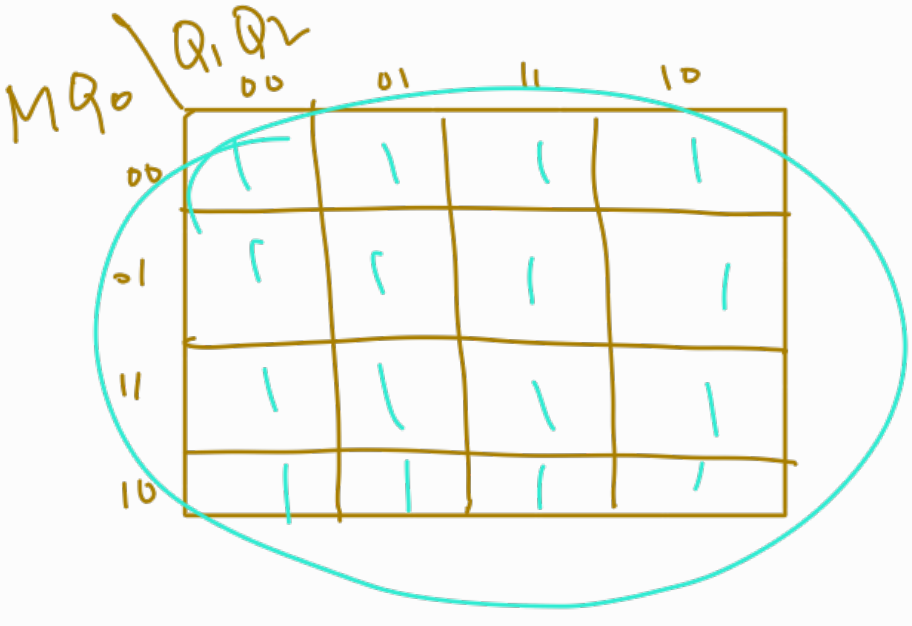


$T_0 = \bar{M} Q_1 Q_2 + M \bar{Q_1} \bar{Q_2}$



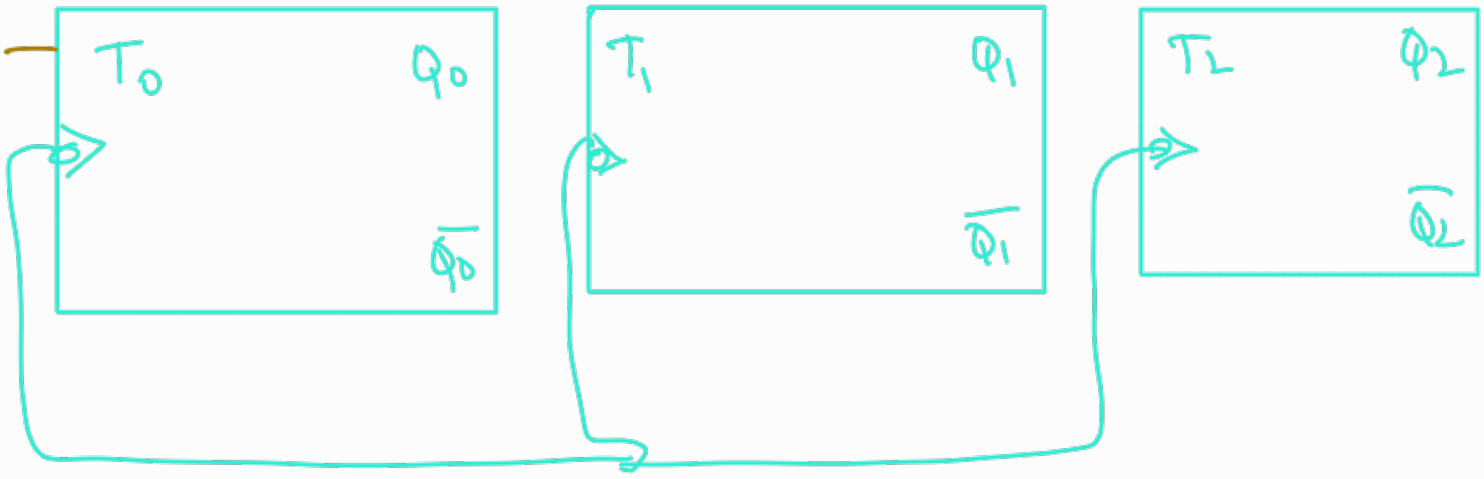
$T_1 = \bar{M} Q_2 + M \bar{Q_2}$

$= M \oplus Q_2$



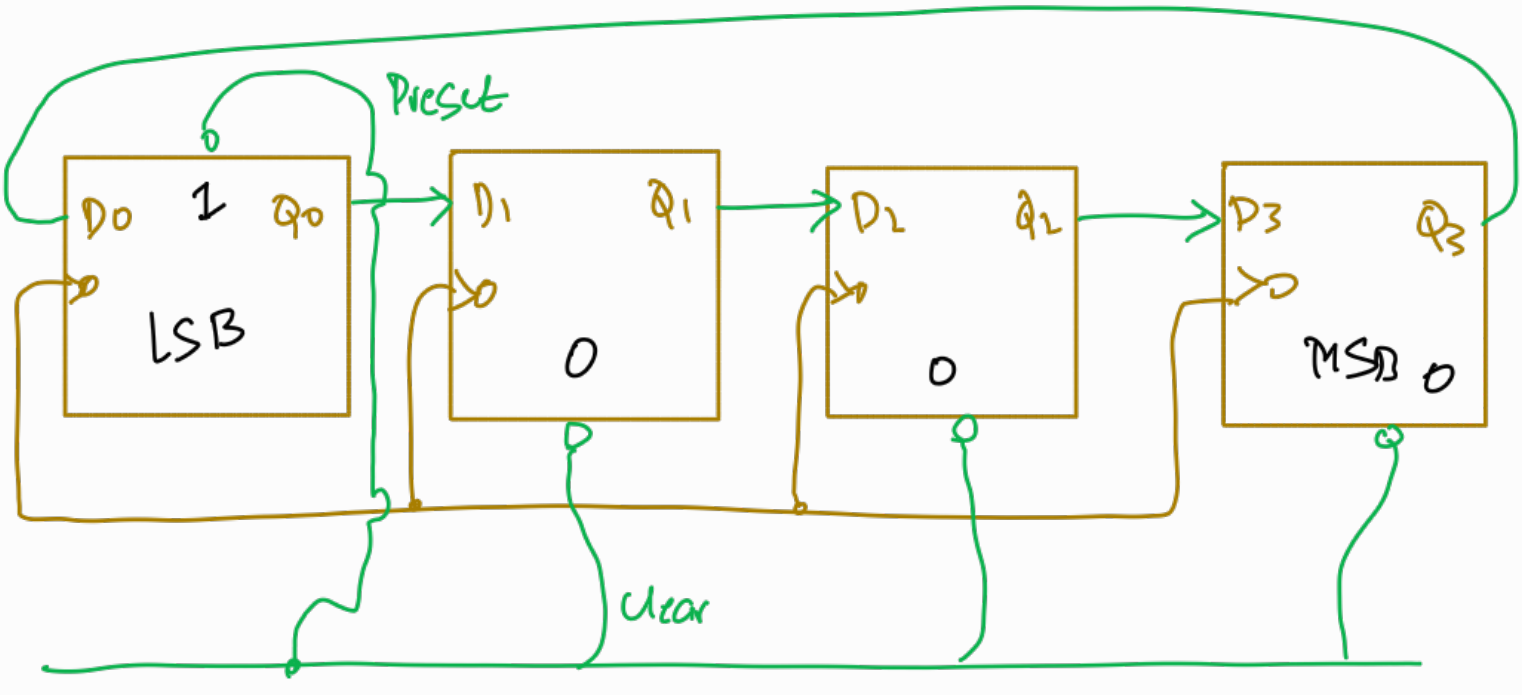
$T_2 = 1$

now just make the complete diagram.



Ring counter:: is a application of Shift Register.

The only change is that the output of Last FF is connected to the input of first FF.



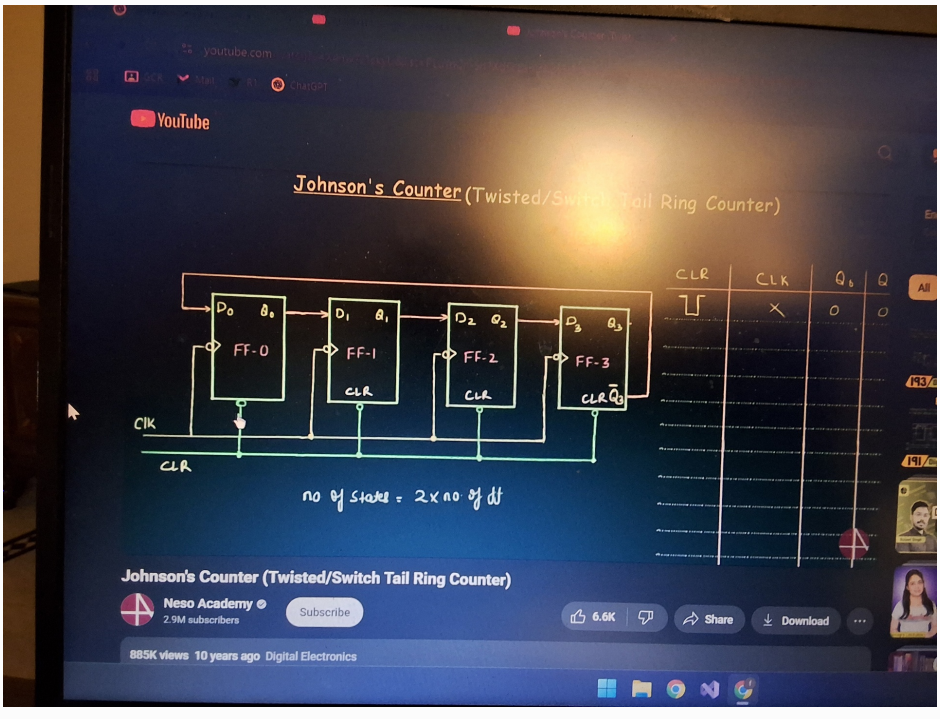
ORI → Active low given.

no. of states = no. of FF used

DRI	CLK	Preset Q_0	Q_1	Q_2	Q_3	State
0	X	1	0	0	0	1
1	↓	0	1	0	0	2
1	↓	0	0	1	0	3
1	↓	0	0	0	1	4
1	↓	1	0	0	0	1

$Q_0 \rightarrow D_1$ $Q_1 \rightarrow D_2$ $Q_3 \rightarrow D_0$

Johnson counter: no of states = 2 x no of ff.



Arbitary counter :- counts Randomly.

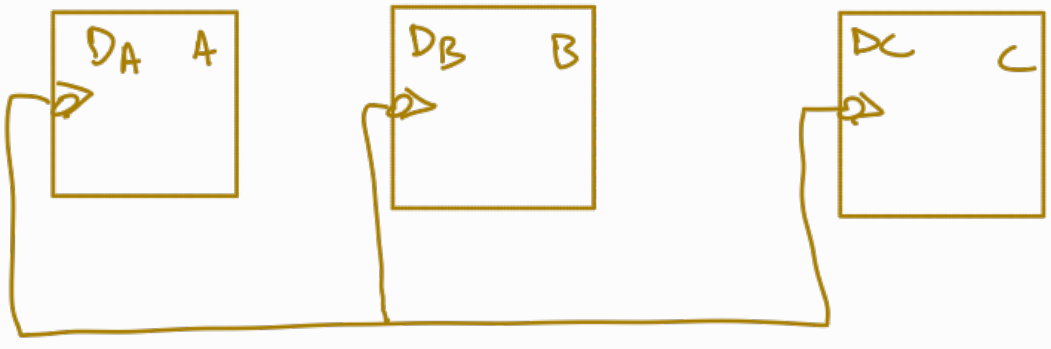
e.g design a counter in the following way →

0, 1, 2, 4, 5, 6, 0

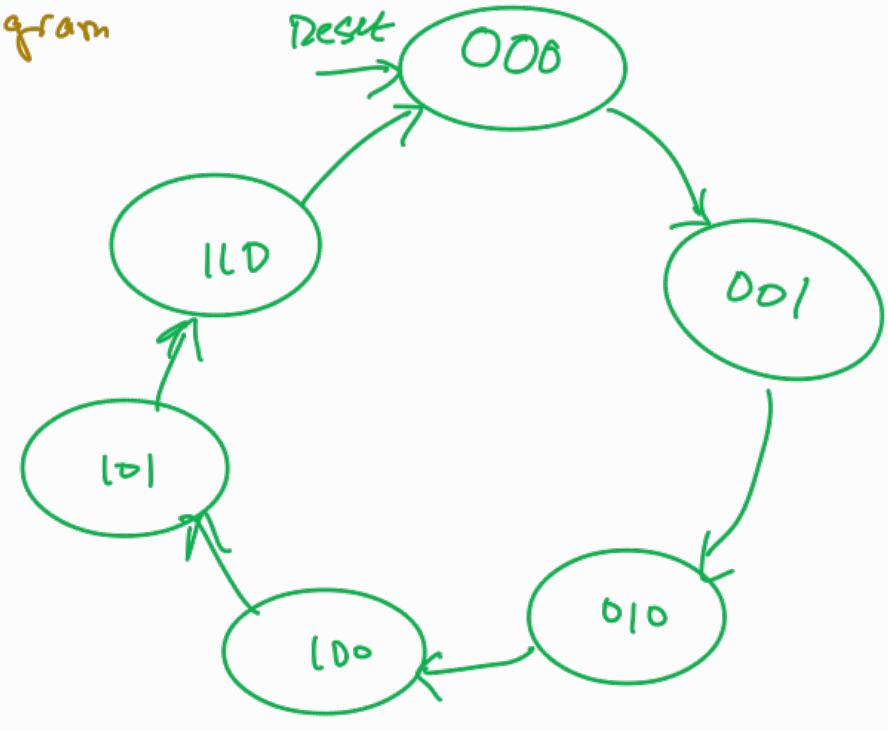
Min bits required = (3) ff.

hence $2^3 = 8$ states count

using D flipflops.



State diagram



(Excitation table DFF)

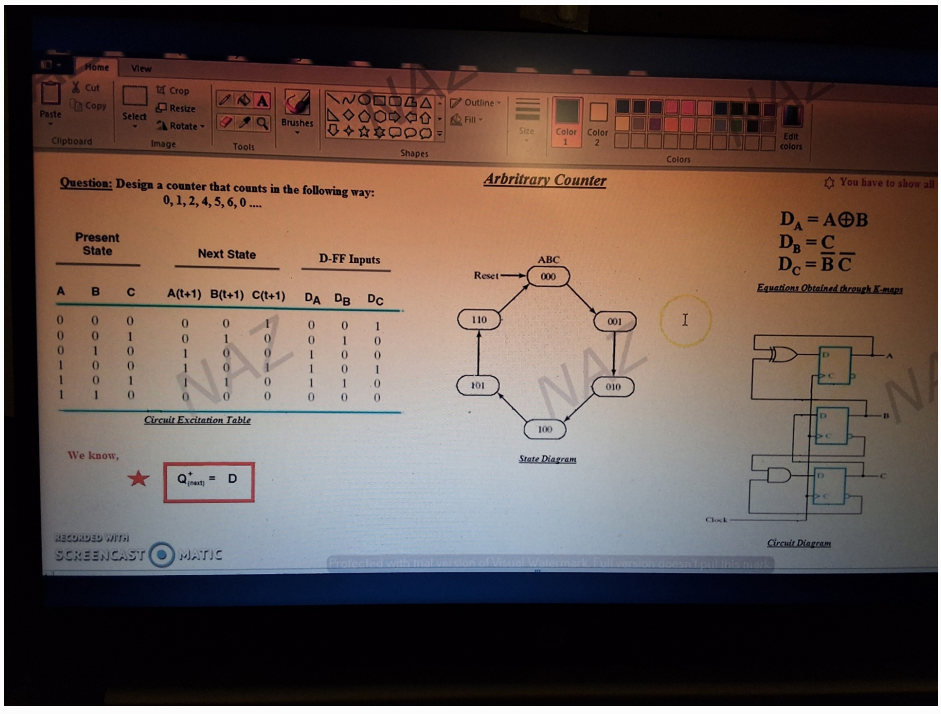
Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

A	B	C	D_A	D_B	D_C
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0



Max counts till 6.

So T.T values must also be six



Question: Design a counter that counts in the following way:
0, 1, 2, 4, 5, 6, 0 ....

Arbitrary Counter

You have to show all

Present State			Next State			D-FF Inputs		
A	B	C	A(t+1)	B(t+1)	C(t+1)	D_A	D_B	D_C
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	0
0	1	0	1	0	0	1	0	0
1	0	0	1	0	1	1	0	1
1	0	1	1	1	0	1	1	0
1	1	0	0	0	0	0	0	0

Circuit Excitation Table

We know,

$Q^{+}_{(next)} = D$

State Diagram

$D_A = A \oplus B$

$D_B = \overline{C}$

$D_C = \overline{B}\,\overline{C}$

Equations Obtained through K-maps

Circuit Diagram

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